CS6847 — Cloud Computing + Distributed Computing

Tier	B
Phase (order)	7
Do this first	CS3500, CS3205, CS3700
Estimated effort	12 weeks
Unlocks	CS6440, CS6851

Roadmap

1. Virtualisation, containers, orchestration.
2. Distributed systems: time, consensus (Raft), replication, fault tolerance.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- MIT 6.824 Distributed Systems (Spring 2020) (20 videos). MIT 6.824 *Distributed Systems* (Spring 2020, Robert Morris, official course channel): MapReduce, Raft, GFS, Spanner. Do the Go labs.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Distributed Systems lecture series (23 videos). Martin Kleppmann *Distributed Systems*: short, rigorous, free lecture notes.
- Cloud Computing and Distributed Systems - Prof. Rajiv Misra (21 videos). NPTEL *Cloud Computing and Distributed Systems* (Rajiv Misra, IIT Patna).
- Big Data Computing.:-Prof. Rajiv Misra (35 videos). NPTEL *Big Data Computing* (Rajiv Misra): Hadoop/Spark/Kafka.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–2	Intro, RPC, threads in Go. Lab 1: MapReduce (pdos.csail.mit.edu/6.824).
2	3–4	GFS, primary-backup replication.
3-4	5–7	Raft. Lab 2: implement Raft (the hardest and most valuable part).
5	8–11	ZooKeeper, CRAQ, Aurora, Frangipani. Lab 3: fault-tolerant key/value store on Raft.
6	12–14	Distributed transactions, Spanner, optimistic concurrency control.

Week	Lectures	What to do
7	15–20	Spark, Memcache at Facebook, causal consistency, certificate transparency, Bitcoin, Blockstack.

▼ **All lectures in *MIT 6.824 Distributed Systems (Spring 2020)* (20 videos)**

1. Lecture 1: Introduction
2. Lecture 2: RPC and Threads
3. Lecture 3: GFS
4. Lecture 4: Primary-Backup Replication
5. Lecture 5: Go, Threads, and Raft
6. Lecture 6: Fault Tolerance: Raft (1)
7. Lecture 7: Fault Tolerance: Raft (2)
8. Lecture 8: Zookeeper
9. Lecture 9: More Replication, CRAQ
10. Lecture 10: Cloud Replicated DB, Aurora
11. Lecture 11: Cache Consistency: Frangipani
12. Lecture 12: Distributed Transactions
13. Lecture 13: Spanner
14. Lecture 14: Optimistic Concurrency Control
15. Lecture 15: Big Data: Spark
16. Lecture 16: Cache Consistency: Memcached at Facebook
17. Lecture 17: COPS, Causal Consistency
18. Lecture 18: Fork Consistency, Certificate Transparency
19. Lecture 19: Bitcoin
20. Lecture 20: Blockstack

Books, notes and websites

- MIT 6.5840 (formerly 6.824) *Distributed Systems* labs + lectures
- Book: Kleppmann, *Designing Data-Intensive Applications*
- Docker and Kubernetes official tutorials

Practice / proof of understanding

Implement Raft (MIT 6.5840 lab 2) and deploy a 3-service app with Docker Compose.

What your IITM course sheet says

From the PDF: CS6847

Official description: NIL

Official syllabus (trimmed):

> Introduction to Cloud Computing: Cluster Computing, Grid Computing, Utility Computing, Cloud Computing,- Concepts of virtualization, Pay-as-you go model Basics of Service-oriented Computing: Web services, Web Service Description Languages, SOAP and REST services, service composition, service match-making, service mashups Public Clouds: concepts of IaaS, PaaS, SaaS, Amazon case study: EC2, S3, Amazon SQS, Simple DB, etc., cloud security Private Clouds: Eucalyptus architecture, Device clouds, Internet of Things (IOT) Hybrid clouds: issues with data-intensive applications, cloud bursting algorithms Working with Big Data: Big Table, Google File System, Hadoop Distributed File System (HDFS), Map-reduce on Hadoop, Elastic Map-reduce in Amazon, Iterative machine-learning applications with Mahout, Iterative/Recursive applications with Generate-Map-Reduce, Graph applications, Dryad, DryadLINQ, HBase, Pig, Hive, etc., key-value stores with relaxed consistency models – Amazon Dynamo, Virat

Official textbooks: NIL

Official references: 1. Hadoop: The Definitive Guide, O'Reilly, 2009 2. Grid Computing – A research Monograph, D. Janakiram, McGraw-Hill 3. Selected research papers

From the PDF: CS6440

Official description: Notion of time in distributed systems; Logical clocks, vector clocks, bitmatrix clocks, virtual clocks; Shared memory vs. message passing; Consensus; Byzantine agreement, agreement protocols, commit protocols; Mutual exclusion in distributed systems; Distributed algorithms; 2, 3 tier c/s architectures; Languages for distributed computing; Linda, CSP, DP, SR; Distributed operating systems; Distributed file systems; Distributed databases; Distributed objects -DCOM, CORBA, Java.

Official textbooks: NIL